



Diesel Generator Set

mtu 12V4000 DS1650

380V – 11 kV/50 Hz/data center continuous power/
NEA (ORDE) optimized/12V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 1490 kVA - 1600 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	260
Model	12V4000G14F		Engine jacket water capacity: l	160
Type	4-cycle		Intercooler coolant capacity: l	40
Arrangement	12V		Combustion air requirements	
Displacement: l	57.2			
Bore: mm	170			
Stroke: mm	210		Combustion air volume: m³/s	1.8
Compression ratio	16.4		Max. air intake restriction: mbar	50
Rated speed: rpm	1500		Cooling/radiator system	
Engine governor	ECU 9			
Max power: kWm	1420			
Air cleaner	dry		Coolant flow rate (HT circuit): m³/hr	56
Fuel system			Coolant flow rate (LT circuit): m³/hr	30
			Heat rejection to coolant: kW	545
			Heat radiated to charge air cooling: kW	260
			Heat radiated to ambient: kW	75
			Fan power for electr. radiator (40°C): kW	55
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	16			
Fuel consumption ²⁾				
	l/hr	g/kwh	Exhaust gas temp. (after turbocharger): °C	505
At 100% of power rating:	342.2	200	Exhaust gas volume: m³/s	4.9
At 75% of power rating:	274.6	214	Maximum allowable back pressure: mbar	85
At 50% of power rating:	200.2	234	Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard)	380 V	1280	1600	2431	1216	1520	2309
	400 V	1280	1600	2309	1216	1520	2194
	415 V	1280	1600	2226	1216	1520	2115
Marathon 743RSL7090 (Low voltage Marathon)	380 V	1272	1590	2416	1224	1530	2325
	400 V	1264	1580	2281	1224	1530	2208
	415 V	1192	1490	2073	1192	1490	2073
Marathon 744RSL7091 (Low voltage Marathon oversized)	380 V	1272	1590	2416	1224	1530	2325
	400 V	1264	1580	2281	1224	1530	2208
	415 V	1192	1490	2073	1192	1490	2073
Marathon 1020FDH7095 (Medium volt. marathon)	11 kV	1264	1580	83	1216	1520	80
Leroy Somer LSA53.2 VL6 (Medium volt. Leroy Somer)	11 kV	1272	1590	83	1216	1520	80

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.